

Geometric Queries on Closed Implicit Surfaces for Walk on Stars

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Abstract

Walk on stars (WoSt) is currently one of the most advanced Monte Carlo solvers for PDEs. Unfortunately, the lack of reliable geometric query approaches has hindered its applicability to boundaries defined by implicit surfaces. This work proposes a geometric query framework over closed implicit surfaces for WoSt, under the scope of *walkin' Robin* [Miller et al. 2024b]. Our key observation is that all WoSt queries can be formulated as constrained global optimization or constraint satisfaction problems. Based on our formulations, to solve the highly non-convex problems, we adopt a branch-and-bound approach based on interval analysis. To the best of our knowledge, our method is the first to study closest silhouette point queries and Robin radius bound queries on closed implicit surfaces. Our formulations and methods first enable mesh-free PDE solving via WoSt when boundaries are defined by closed implicit surfaces.

CCS Concepts

• Computing methodologies → Shape analysis.

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1 Introduction

Partial differential equations (PDEs) are fundamental in modeling physical phenomena and widely used in graphics. Recently, Monte Carlo solvers for PDEs have gained increasing attention, with *walk on stars (WoSt)* [Miller et al. 2024b; Sawhney et al. 2023], an extension of *walk on spheres (WoS)* [Muller 1956], emerging as a promising approach. Unlike traditional methods, WoSt avoids discretization and thus bypasses the robustness and performance challenges of volumetric meshing, making it well-suited for increasingly complex geometries in graphics.

Despite these advantages, WoSt has not yet been demonstrated to operate entirely in domains defined by implicit surfaces (hereafter referred to as *implicit domains*). Implicit surfaces can represent intricate geometric details, support robust Boolean/blending operations, and are central to many modern scene representations. A volumetric PDE solver supporting implicit domains directly would

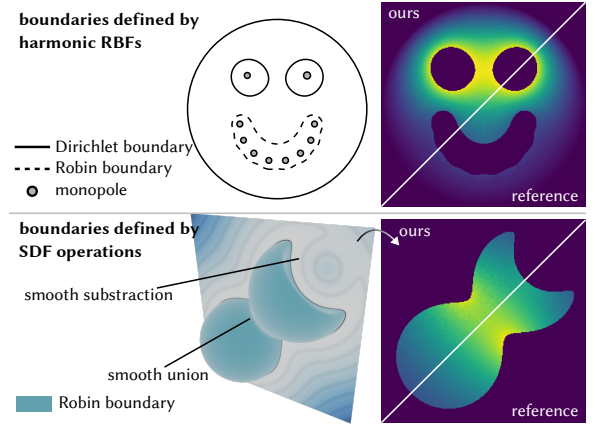


Figure 1: Walk on stars (WoSt) with boundaries defined by closed implicit surfaces. Our method can perform complete WoSt geometric queries on various closed implicit representations (here we demonstrate harmonic RBFs [Miller et al. 2024a] and SDF operations). WoSt based on our method yields correct solutions to the Laplace Dirichlet-Robin problems.

inherit these benefits, but current WoSt implementations remain tied to boundary meshes or only support Dirichlet boundaries defined by implicit surfaces [Miller et al. 2024a], limiting its strengths in implicit domains. The difficulty in extending WoSt to implicit domains is that, while WoS only requires closest point queries, WoSt additionally requires geometric queries that have not been well studied on implicit surfaces. To facilitate explanation, we list all the geometric queries required by WoSt:

- (1) Closest point query (CPQ).
- (2) Ray intersection query.
- (3) Closest silhouette point query (CSPQ).
- (4) Robin radius bound query (RRBQ).
- (5) Point sampling on *reflecting* (Neumann/Robin) boundaries.

While queries (1), (2), and (5) are well studied, no existing methods address (3) and (4) on implicit surfaces. For instance, silhouette detection has been studied in non-photorealistic [Bremer and Hughes 1998] and differentiable [Zhou et al. 2024] rendering, but methods for CSPQ remain unexplored on implicit surfaces; and RRBQ has only been studied for boundary meshes [Miller et al. 2024b]. Consequently, WoSt still falls back to intermediate boundary meshing to handle implicit domains, undermining its mesh-free appeal and degrading geometric fidelity.

In response, we propose a geometric query solution over closed (compact and boundary-free) implicit surfaces for WoSt, corresponding



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to PDE problems where the Dirichlet and reflecting boundaries are defined by closed implicit surfaces. Our approach builds on an observation: all the queries above can be formulated as constrained global optimization or constraint satisfaction problems. Based on our proposed formulations, to run optimization and satisfaction on the non-convex function landscape, we adopt branch-and-bound techniques based on interval analysis (IA) [Snyder 1992]. From a contributions perspective, we are the first to study the formulation and implementation of queries (3) and (4) on closed implicit surfaces, and the first to realize a complete WoSt when boundaries are defined by closed implicit surfaces.

2 Preliminary: Interval Analysis (IA) for Constrained Global Optimization and Constraint Satisfaction

The geometric queries exhibit non-convex landscapes, and WoSt often admits conservative queries. Consequently, we adopt IA as the tool for our queries. We recall two relevant methods introduced by Snyder [1992]: MINIMIZE for constrained global optimization, and SOLVE for constraint satisfaction. In the methods, *boxes*, i.e., Cartesian product of intervals, are used to divide the space. For an *objective function* g , its corresponding *objective inclusion function* $\square g$ maps a box Y to an interval enclosing all $g(x), \forall x \in Y$. *Constraint inclusion function* $\square G$ that takes a box Y returns a three-value logic:

- positive: constraint G holds for all points in Y ,
- negative: constraint G fails for all points in Y ,
- unknown: both satisfying and unsatisfying points exist in Y .

Solution acceptance set constraint $\square A$ returns true if the search box is fine enough (that meets the specified tolerance). X is the initial box. The algorithms are as follows:

Listing 1: MINIMIZE: Constrained Global Optimization

```

1 def MINIMIZE(  $\square g$ ,  $\square G$ ,  $\square A$ ,  $X$  ):
2   L  $\leftarrow$  PriorityQueue({X})
3   UB  $\leftarrow +\infty$  # upper bound for pruning
4   results  $\leftarrow \emptyset$  # multiple solutions may arise
5   while L  $\neq \emptyset$ :
6     Y  $\leftarrow$  L.pop_min_lower_bound() # best candidate
7     if  $\square A(Y)$  = true: # stricter tests may be applied
8       results  $\leftarrow$  results  $\cup$  {Y} # accept Y
9     else:
10      for Y_i in subdivide(Y):
11        if  $\square G(Y_i)$  = negative:
12          continue # prune constraint-violating boxes
13        [a_i, b_i]  $\leftarrow$   $\square g(Y_i)$ 
14        if a_i > UB:
15          continue # prune suboptimal boxes
16        L.insert(Y_i, priority=a_i)
17        UB  $\leftarrow$  min(UB, b_i) # update upper bound
18  return results

```

Listing 2: SOLVE: Constraint Satisfaction

```

1 def SOLVE(  $\square G$ ,  $\square A$ ,  $X$  ):
2   L  $\leftarrow$  {X}, results  $\leftarrow \emptyset$ 
3   while L  $\neq \emptyset$ :
4     Y  $\leftarrow$  remove(L) # take a box from L
5     if  $\square G(Y)$  = positive  $\wedge$   $\square A(Y)$  = true:
6       results  $\leftarrow$  results  $\cup$  {Y} # accept fine &
7         constraint-satisfying boxes
8     elif  $\square G(Y)$  = unknown:
9       L  $\leftarrow$  L  $\cup$  subdivide(Y) # further subdivide unknown boxes
10  return results

```

In our paper, SOLVE is used for point sampling on reflecting boundaries (section 3.4), while the rest utilizes MINIMIZE.

3 Method

We consider a domain $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) whose boundary¹ is partitioned into two closed implicit surfaces: the Dirichlet boundary $\partial\Omega_D$ and the reflecting boundary $\partial\Omega_R$. They are defined as the zero level sets of smooth implicit functions $f_D, f_R : \mathbb{R}^d \rightarrow \mathbb{R}$, with the regularity condition $\nabla f_D(x) \neq 0, \forall x \in \partial\Omega_D$ and $\nabla f_R(z) \neq 0, \forall z \in \partial\Omega_R$. Under this condition, ∇f_D and ∇f_R are parallel to the corresponding surface normals. No Lipschitz assumption is required. We denote by $x \in \Omega$ a query point, by $y \in \partial\Omega_D$ a Dirichlet boundary point, and by $z \in \partial\Omega_R$ a reflecting boundary point. The closed and open balls centered at x with radius R are written as $B(x, R)$ and $\bar{B}(x, R) = B(x, R) \setminus \partial B(x, R)$, resp. Some notations here differ from those in the WoSt paper for contextual consistency; readers unfamiliar with WoSt may refer to the supplementary material.

3.1 Closest Point Query (CPQ) & Ray Intersection Query

In WoSt, CPQ is used on $\partial\Omega_D$ and ray intersection query is used on $\partial\Omega_R$, these queries are well-studied as global optimization problems; for completeness, we briefly summarize them here. The CPQ can be formulated as:

$$y_D = \arg \min_{y \in \mathbb{R}^d} \|y - x\|^2 \quad \text{subject to} \quad f_D(y) = 0, \quad (1)$$

and ray intersection query can be formulated as:

$$z_I = \arg \min_{z \in \mathbb{R}^d} \|z - x\|^2 \quad \text{subject to} \quad f_R(z) = 0, \quad z \in \{x + tv \mid t > 0\}, \quad (2)$$

where $v \in \mathbb{R}^d$ is a unit direction vector. Apart from using MINIMIZE, many efficient methods exist, which we do not discuss further here. CPQ yields the *Dirichlet distance* $R_D = \|y_D - x\|$.

3.2 Closest Silhouette Point Query (CSPQ)

A silhouette point is a point whose normal is perpendicular to the viewing direction. In WoSt, CSPQ is used on $\partial\Omega_R$ to produce the *silhouette bound* R_S for the radius of the star-shaped region such that no silhouette exists within the star-shaped region. We propose to formulate CSPQ as the following problem when $x \notin \partial\Omega_R$:

$$z_S = \arg \min_{z \in \mathbb{R}^d} \|z - x\|^2 \quad \text{subject to} \quad \begin{cases} f_R(z) = 0 \\ \nabla f_R(z) \cdot (z - x) = 0, \end{cases} \quad (3)$$

therefore we have $R_S = \|z_S - x\|$. In WoSt, R_S should satisfy $R_S \leq R_D$, thus we introduce the additional constraint $\|z - x\| \leq R_D$. To narrow the search space, we initialize the MINIMIZE algorithm with a box centered at x and of side length $2R_D$. When the problem involves Robin boundaries, we slightly shrink R_S to ensure $\cos \theta(z) = \frac{n_z \cdot (z - x)}{r} = \frac{|\nabla f_R(z) \cdot (z - x)|}{\|\nabla f_R(z)\| r} > 0, \forall z \in B(x, R_S) \cap \partial\Omega_R^2$, thereby avoiding singularities in *reflectance* ρ_μ [Miller et al. 2024b,

¹In this section, "boundary" refers to the boundary of the PDE problems and should be distinguished from the boundaries of the implicit surfaces discussed in the context of open and closed surfaces.

² ∇f_R always points outward from the implicit surface, irrespective of x 's location (inside or outside); thus we take the absolute value in the dot product.

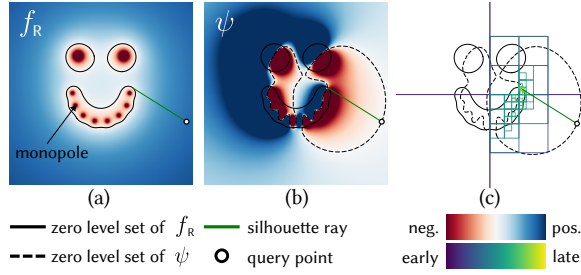


Figure 2: Closest silhouette point query (CSPQ). Figures (a) and (b) illustrate the landscapes of f_R and $\psi = \nabla f_R(z) \cdot (z - x)$, *resp.* Figure (c) visualizes the minimization process.

eqs. 9 and 33]. The query strategy for $x \in \partial\Omega_R$ is discussed in the supplementary material.

3.3 Robin Radius Bound Query (RRBQ)

In walkin' Robin, to ensure $\rho_\mu \in [0, 1]$ for a stable Monte Carlo estimator, Miller et al. [2024b] propose shrinking R_S to the *Robin radius bound* $R_R \leq R_S$. We now discuss the query for R_R by dimension.

3.3.1 2D Case. In 2D, a desired R_R ensures, $\forall z \in B(x, R_R) \cap \partial\Omega_R \subset B(x, R_S) \cap \partial\Omega_R$, the following holds [Miller et al. 2024b, eq. 35]:

$$R_R \leq r \exp\left(\frac{\cos \theta}{\mu(z)r}\right), \quad (4)$$

where $r = r(z) = \|z - x\|$, $\cos \theta > 0$ (section 3.2), and $\mu(z) > 0$ is the prescribed Robin coefficient defined on $\partial\Omega_R$. Since z appears entirely on the right-hand side of eq. (4), it is natural to consider taking the minimum of it over $B(x, R_S) \cap \partial\Omega_R$ as the upper bound for R_R , which corresponds to the following optimization problem:

$$R_R^* = \min_{z \in \mathbb{R}^2} r \exp\left(\frac{\cos \theta}{\mu(z)r}\right) \quad \text{subject to} \quad \begin{cases} f_R(z) = 0 \\ r \leq R_S. \end{cases} \quad (5)$$

We further find that the bound R_R^* here is tight, *i.e.*, there exists no $\tilde{R}_R > R_R^*$ satisfying eq. (4), because $\forall z \in B(x, R_S) \cap \partial\Omega_R$ we have:

$$r \exp\left(\frac{\cos \theta}{\mu(z)r}\right) - r = r \left[\exp\left(\frac{\cos \theta}{\mu(z)r}\right) - 1 \right] > 0. \quad (6)$$

Therefore, the minimizer z^* necessarily satisfies $r^*(z^*) < R_R^*$, or equivalently $z^* \in \hat{B}(x, R_R^*) \cap \partial\Omega_R$, so $\tilde{R}_R > R_R^*$ would necessarily violate eq. (4) at z^* . Hence, R_R^* is guaranteed to be the tight bound.

Finally, we take the lower bound of the interval for R_R^* for numerical stability. This also applies to the 3D case.

3.3.2 3D Case. In 3D, eq. (4) takes a different form [Miller et al. 2024b, eq. 11]:

$$R_R \leq \frac{r}{1 - \frac{\cos \theta}{\mu(z)r}} \quad \text{when} \quad r > \frac{\cos \theta}{\mu(z)}. \quad (7)$$

The definition of symbols follow the 2D case except that the points and vectors are defined in $\mathbb{R}^3$. The biggest difference lies in the condition $r > \frac{\cos \theta}{\mu(z)}$. When $0 < r \leq \frac{\cos \theta}{\mu(z)}$, R_R is no longer subject

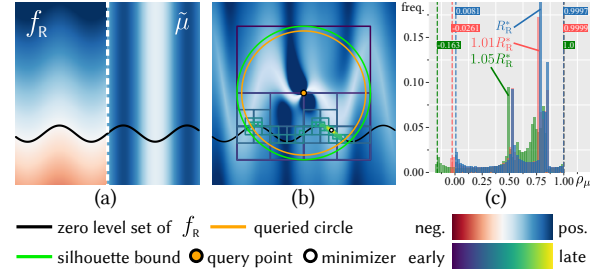


Figure 3: Robin radius bound query (RRBQ). Figure (a) shows the landscape of f_R and $\tilde{\mu}$ (section 3.3.3). Figure (b) shows the landscape of the objective, optimization process and result. In figure (c), we randomly sample points on $\partial\Omega_R$ inside the queried circle, and plot the distribution of ρ_μ [Miller et al. 2024b, eqs. 9 and 33]. As illustrated, $\rho_\mu \in [0, 1]$ when the radius equals to R_R^* , while increasing the radius slightly yields invalid ranges of ρ_μ .

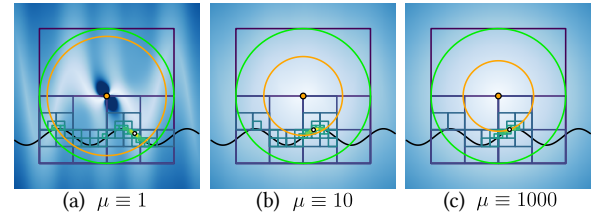


Figure 4: Sanity check on the effect of increasing μ on the radius. We set $\tilde{\mu}(z)$ to a constant *w.r.t.* z , increase it from (a) to (c), and perform RRBQ. The results show the query converges to CPQ on $\partial\Omega_R$ as μ increases, which agrees with the results reported by Miller et al. [2024b, fig. 7].

to eq. (7), and it can be made arbitrarily large [Miller et al. 2024b, section 4.2.1]. We also observe that eq. (7) naturally guarantees:

$$\frac{r}{1 - \frac{\cos \theta}{\mu(z)r}} - r = \frac{\frac{\cos \theta}{\mu(z)}}{1 - \frac{\cos \theta}{\mu(z)r}} > 0 \quad \text{when} \quad r > \frac{\cos \theta}{\mu(z)}. \quad (8)$$

Thus, the following optimization problem also yields a tight bound:

$$R_R^* = \min_{z \in \mathbb{R}^3} \frac{r}{1 - \frac{\cos \theta}{\mu(z)r}} \quad \text{subject to} \quad \begin{cases} f_R(z) = 0 \\ \frac{\cos \theta}{\mu(z)} < r \leq R_S. \end{cases} \quad (9)$$

We denote the objective function in eq. (9) as $O(z; x)$. Using IA to impose $r > \cos \theta / \mu(z)$ may be overkill, as in condition-violating regions, we no longer need to evaluate its range. We instead adopt a simpler approach: based on the observation that $O(z; x) \rightarrow +\infty$ as $r \rightarrow (\cos \theta / \mu(z))^+$, we set a new optimization objective:

$$\tilde{O}(z; x) = \begin{cases} \frac{r}{1 - \frac{\cos \theta}{\mu(z)r}}, & r > \frac{\cos \theta}{\mu(z)} \\ +\infty, & \text{o.w.} \end{cases} \quad (10)$$

Since we are performing a minimization, the optimizer will automatically bypass the $+\infty$ region. This objective can be computed

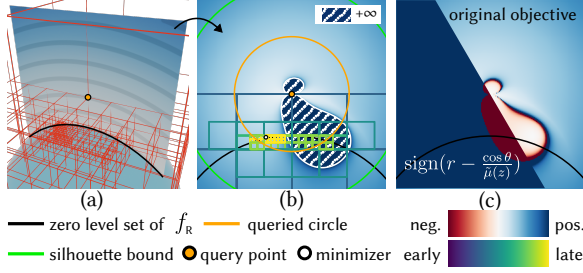


Figure 5: RRBQ in 3D. Figures (a) and (b) show the optimization process and results in $\mathbb{R}^3$ and on a slice, *resp.* Comparing (b) and (c), we see that regions where $r > \frac{\cos \theta}{\mu(z)}$ is not satisfied are set to $+\infty$ in eq. (10), then the optimizer skips these regions during the coarse stage. The sampled values (see fig. 3 caption) of ρ_μ fall within $[0.4891, 0.99996]$.

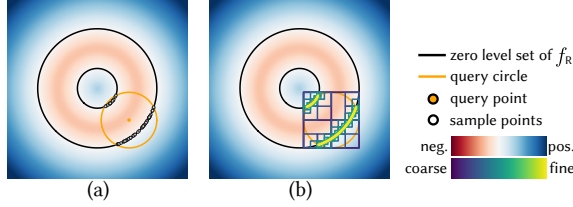


Figure 6: Point sampling on reflecting boundaries. Figure (a) shows the implicit function landscape, the query circle, and the sampled points. Figure (b) illustrates the search boxes.

numerically by clamping the interval of $1 - \frac{\cos \theta(z)}{\mu(z)r}$ to $[0, +\infty)$ (we define division by 0 to yield $+\infty$). In the end we have:

$$R_R^* = \min_{z \in \mathbb{R}^3} \tilde{O}(z; x) \quad \text{subject to} \quad \begin{cases} f_R(z) = 0 \\ r \leq R_S. \end{cases} \quad (11)$$

3.3.3 Discussion & Implementation. By definition, the domain of μ is $\partial\Omega_R$, whereas the IA operates in $\mathbb{R}^d$. Therefore, we extend μ to $\mathbb{R}^d$ by constructing $\tilde{\mu} : \mathbb{R}^d \rightarrow \mathbb{R}$ that satisfies $\tilde{\mu}(z) = \mu(z), \forall z \in \partial\Omega_R$. We initialize the MINIMIZE algorithm with a box centered at x with side length $2R_S$.

3.4 Point Sampling on Reflecting Boundaries

In WoSt, we need to sample points on $\Gamma = B(x, R_R) \cap \partial\Omega_R$ [Sawhney et al. 2023, section 5.2], which can be formalized as the following constraint satisfaction problem:

$$\Gamma := \{z \in \mathbb{R}^d \mid f_R(z) = 0, \|z - x\| \leq R_R\}. \quad (12)$$

We adopt the SOLVE algorithm. The initial search domain is the box centered at x and of side length $2R_R$. In practice, *reservoir sampling* or a cached tree structure can be used to improve performance.

4 Results and Discussion

We implement the method in Julia [Bezanson et al. 2017] using the IntervalArithmetic.jl [Sanders and Benet 2014] library, extending it with forward-mode automatic differentiation via dual

intervals to obtain gradients. We set the tolerance in section 2 to $1/10$ of the ϵ -shell. For each newly introduced query, we conduct verification experiments in figs. 2 to 6. Based on our method, we also use WoSt to solve Laplace equations, as shown in fig. 1. The results match the expected behavior. We provide detailed setup in the supplementary document.

5 Conclusion, Limitations and Future Work

By proposing formulations and adopting IA-based global optimization and constraint satisfaction methods, our present method first enables all necessary geometric queries on closed implicit surfaces for WoSt, enabling mesh-free PDE solving via WoSt in implicit domains. Our work comes with limitations: although our method is, in principle, applicable to neural SDFs [Sharp and Jacobson 2022], we have not implemented this extension; our current boundary sampling strategy is relatively inefficient—recent advances [Ling et al. 2025] may be combined to improve performance; our work primarily serves as a proof-of-concept, leaving high-performance implementations for future work. At present, optimizing implicit surfaces with reflecting boundaries remains an open inverse problem in the Monte Carlo for PDEs community, and we believe our method could serve as a tool toward this goal.

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